

Normal Distribution

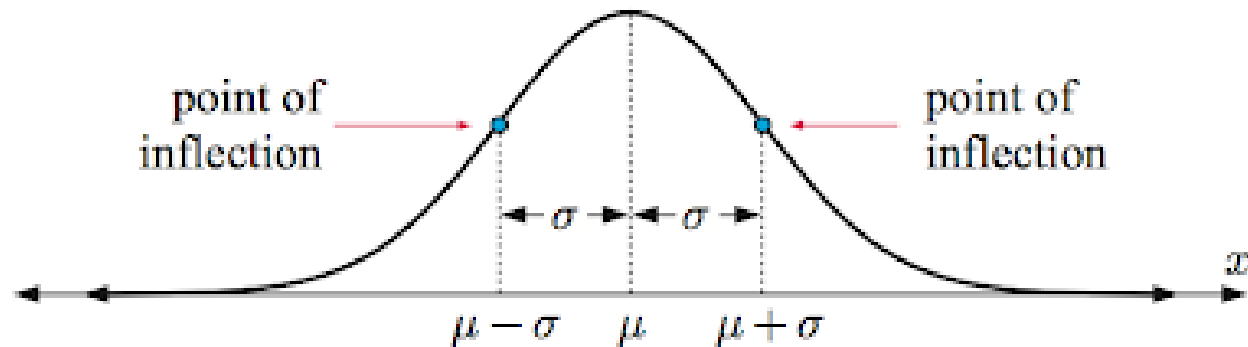
- A continuous random variable X with mean μ and standard deviation σ is said to follow normal distribution if its probability density function is given

by
$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad ; -\infty < x < \infty.$$

- If X follows normal distribution then it is denoted by $X \sim N(\mu, \sigma)$.

Properties of Normal Distribution

- The mean, median and mode are same and it appears at the maximum point($x = \mu$) of the curve. And the max value $f(x = \mu) = \frac{1}{\sigma\sqrt{2\pi}}$.
- The curve is symmetrical about $x = \mu$.
- The curve has a point of inflection points $x = \mu \pm \sigma$.



Properties of Normal Distribution

- Approximately 68% of values in the distribution are within 1 SD of the mean, i.e., above or below.

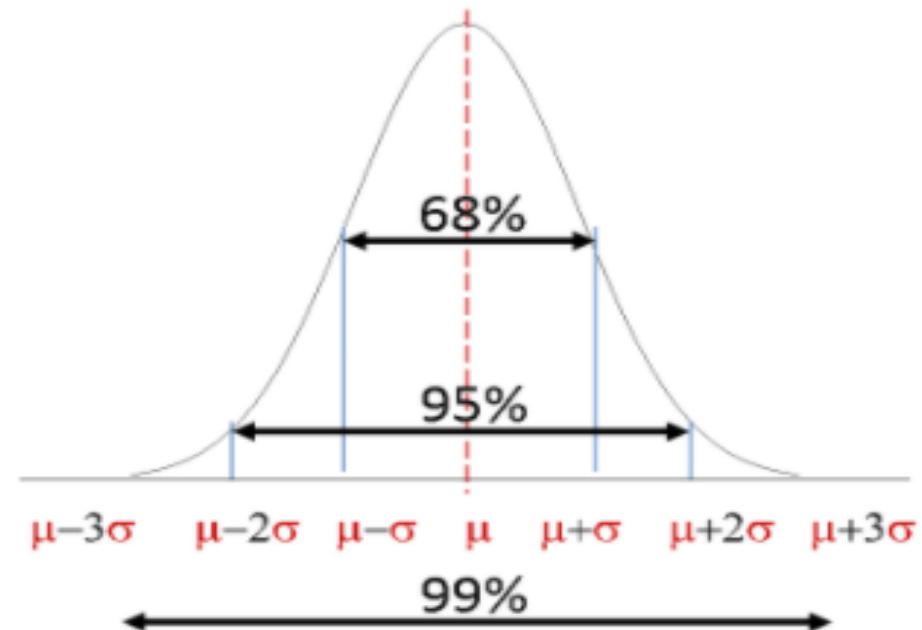
$$P(\mu - \sigma < X < \mu + \sigma) = 0.68$$

- Approximately 95% of values in the distribution are within 2 SD of the mean.

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.95$$

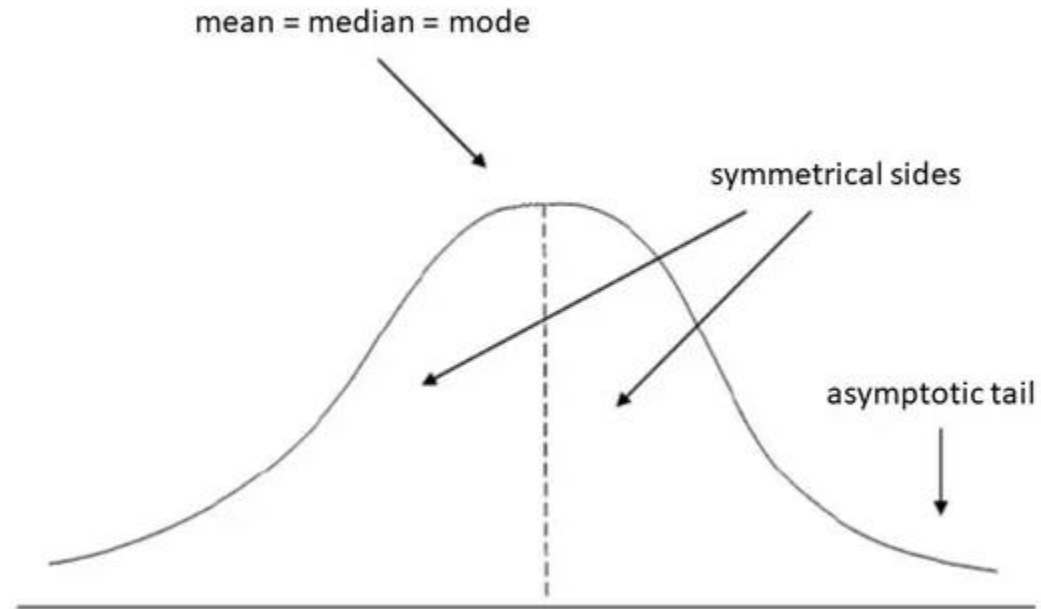
- Approximately 99% of values in the distribution are within 3 SD of the mean.

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.99$$



Properties of Normal Distribution

- The total area under the curve is 1. that is, $\int_{-\infty}^{\infty} f(x)dx = 1$.
- The tails are asymptotic, which means that they approach but never quite meet the horizon.



Properties of Normal Distribution

- The normal distribution is often called the bell curve because the graph of its probability density looks like a bell.
- It is also known as called Gaussian distribution, after the German mathematician Carl Gauss who first described it.
- The normal distribution is the most important probability distribution in statistics because many continuous data in nature and psychology displays this bell-shaped curve when compiled and graphed.

Properties

- The expected value : $E(x) = \mu$.
- The variance : $var(x) = \sigma^2$.
- The moment generating function :

$$M_X(t) = E(e^{tX}) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}, -\infty < t < \infty$$

Transformation of normal random variable to a standard normal variable.

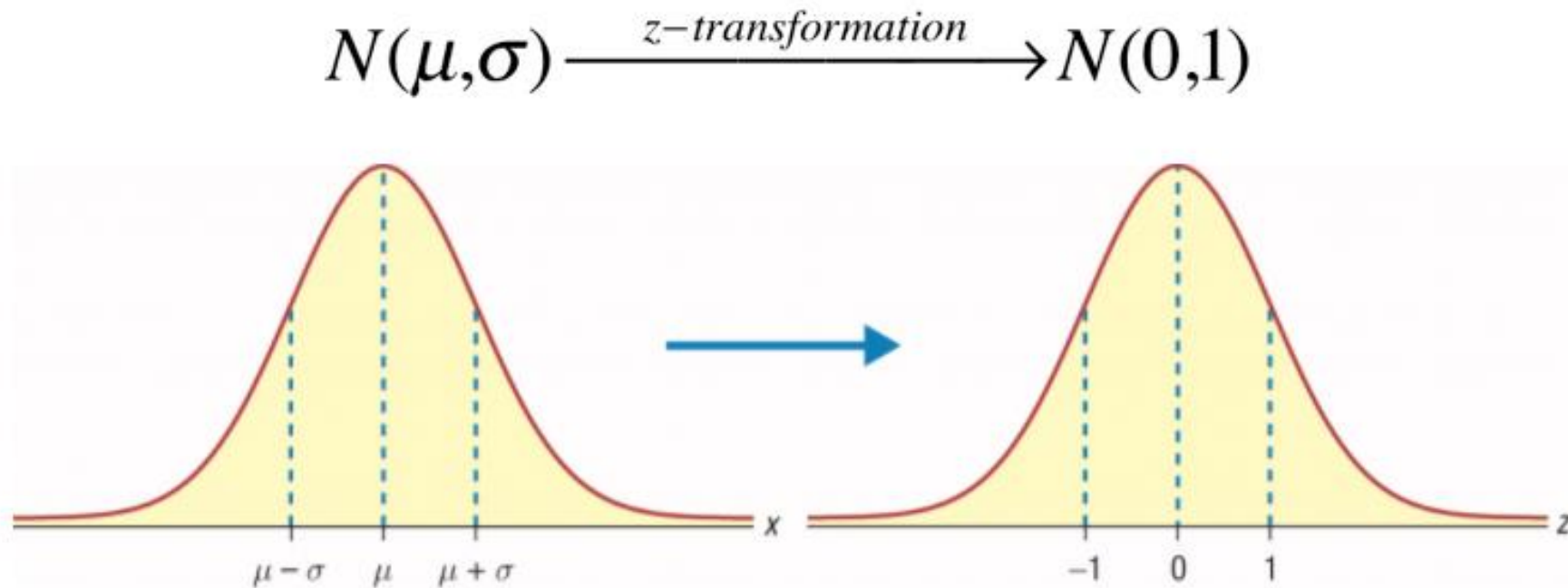
- We are able to transform all the observation of the normal random variable X with mean μ and variance σ^2 into a new set of observations of a normal random variable Z with mean 0 and variance 1 by using the transformation, $Z = \frac{x - \mu}{\sigma}$
- We denote Z is a standard normal random variable.

Transformation of normal random variable to a standard normal variable.

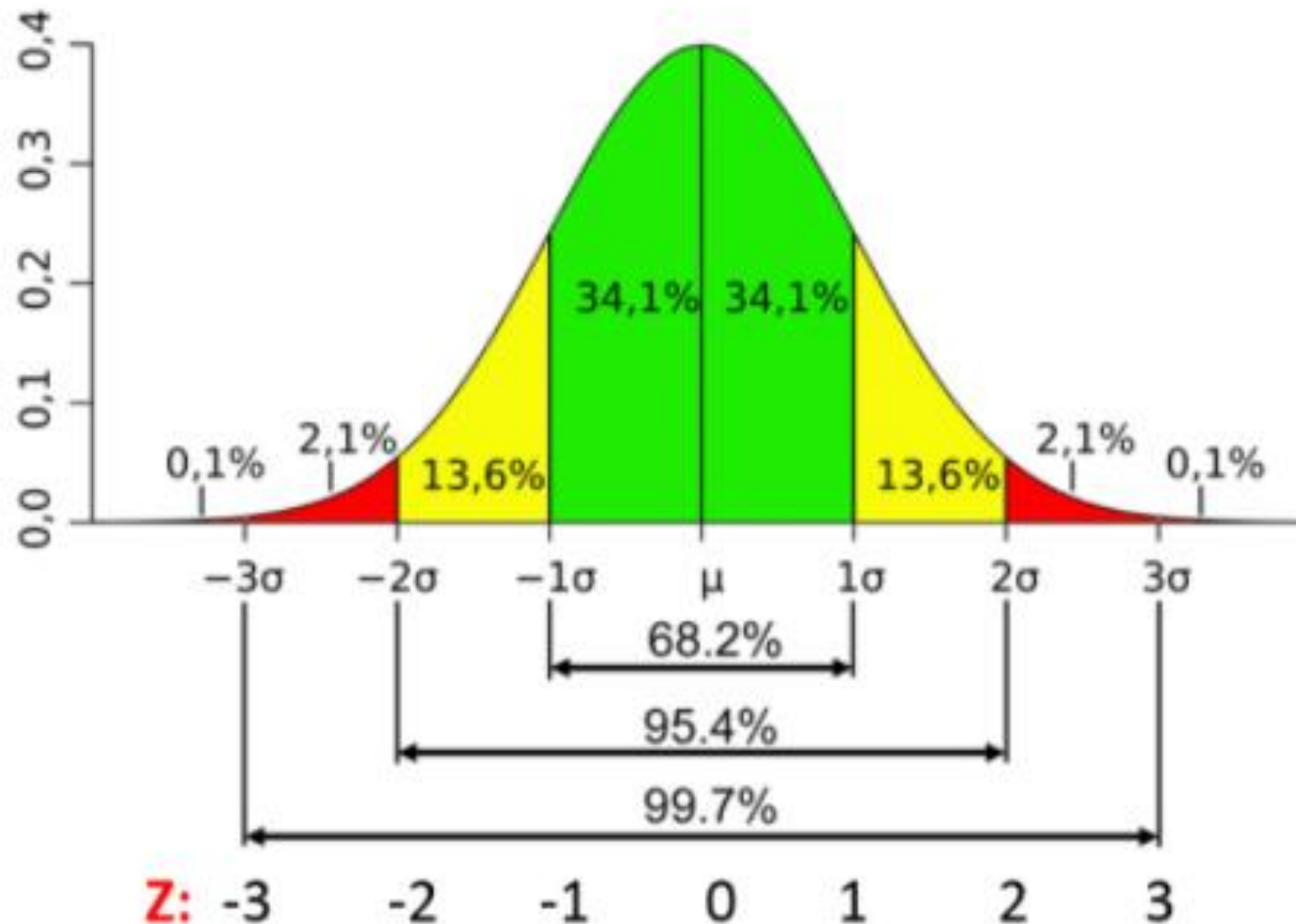
- The probability function of Z is obtained from the pdf of X by $\theta(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}, -\infty < z < \infty$.
- Whenever $P(x_1 \leq x \leq x_2) = \int_{x_1}^{x_2} f(x) dx =$
$$\int_{x_1}^{x_2} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = \frac{1}{\sqrt{2\pi}} \int_{z_1}^{z_2} e^{-\frac{1}{2}z^2} dz = P(z_1 \leq$$

Standard normal variable.

- The standard normal random variable is also denoted by $Z \sim N(0,1)$.

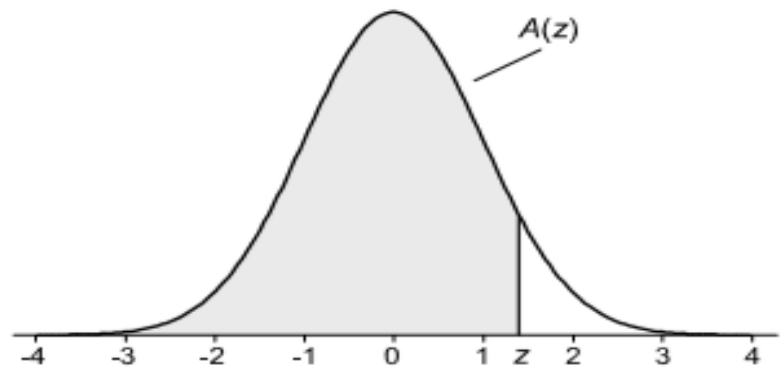


Standard normal variable.



- The table for normal distribution with mean 0 and s.d 1 is used to find the area.

Cumulative Standardized Normal Distribution



$A(z)$ is the integral of the standardized normal distribution from $-\infty$ to z (in other words, the area under the curve to the left of z). It gives the probability of a normal random variable not being more than z standard deviations above its mean. Values of z of particular importance:

z	$A(z)$	
1.645	0.9500	Lower limit of right 5% tail
1.960	0.9750	Lower limit of right 2.5% tail
2.326	0.9900	Lower limit of right 1% tail
2.576	0.9950	Lower limit of right 0.5% tail
3.090	0.9990	Lower limit of right 0.1% tail
3.291	0.9995	Lower limit of right 0.05% tail

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441

Problems

- **Problem 1:** Given a random variable X that follows normal distribution with mean 50 and standard deviation 10. Find the probability that X assumes the value (i) between 45 and 62, (ii) less than 42, (iii) greater than 57.56, (iv) $|X| < 30$. Also do the following (v) find k , if $P(Z > k) = 0.3015$ (vi) find m , if $P(m < Z < -0.18) = 0.4197$. (vii) find s , if $P(X < s) = 0.88877$ (viii) find t , if $P(52 < X < t) = 0.37617$.

- Ans :
- Given $X \sim N(50, 10)$.
- Convert $Z \sim N(0, 1)$ by $Z = \frac{x-50}{10}$.
- Q1 ; $P(45 < X < 62) = P\left(\frac{45-50}{10} < Z < \frac{62-50}{10}\right)$
- $P(-0.5 < Z < 1.2)$

- $P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5)$
- $= 0.8849 - 0.3085 = 0.5764$
- `> pnorm(62,50,10)-pnorm(45,50,10)`
- `[1] 0.5763928`

- Q2 ; $P(X < 42) = P\left(Z < \frac{42-50}{10}\right) = P(Z < -0.8)$

- $P(Z < -0.8) = 0.21186$

- `> pnorm(42,50,10)`

- `[1] 0.2118554`

- Q2 ; $P(X < 42) = P\left(Z < \frac{42-50}{10}\right) = P(Z < -0.8)$

- $P(Z < -0.8) = 0.21186$

- `> pnorm(42,50,10)`

- `[1] 0.2118554`

- Q3 ; $P(X > 57.56) = P\left(Z > \frac{57.56-50}{10}\right) = P(Z > 0.756)$

- $P(Z > 0.756) = 1 - P(Z < 0.756) = 1 - 0.7750 \approx 0.225$

- `> pnorm(57.56,50,10)`

- `[1] 0.7751754`

- `> 1-pnorm(57.56,50,10)`

- `[1] 0.2248246`

- Q4 ; $P(|X| < 30) = P(-30 < X < 30)$
- $P(-30 < X < 30) = P(-8 < X < -2)$

- $P(Z < -2) = 0.0228$
- `> pnorm(-2,0,1)`
- `[1] 0.02275013`
- `> pnorm(30,50,10)`
- `[1] 0.02275013`
- `> pnorm(30,50,10)-pnorm(-30,50,10)`
- `[1] 0.02275013`

- Q5 ;find k , if $P(Z > k) = 0.3015$.
- $1 - P(Z < k) = 1 - 0.3015 = 0.6985$
- From standard normal table, $K=0.52$.

- Suppose,

- Q6 ; find m , if $P(m < Z < -0.18) = 0.4197$.
- $P(Z < -0.18) - P(Z < m) = 0.4197$
- $0.4286 - P(Z < m) = 0.4197$
- $P(Z < m) = 0.0089$
- From table, $m = -2.37$

- Q7 ; find s , if $P(X < s) = 0.88877$.
- $P\left(Z < \frac{s-50}{10}\right) = 0.88877$. let $\frac{s-50}{10} = z_1$
- From standard normal table, $z_1=1.22$.
- Hence $s=62.2$

- Q8 ; find t , if $P(52 < X < t) = 0.37617$.
- $P(X < t) - P(X < 52) = 0.37617$
- $P\left(Z < \frac{t-50}{10}\right) - P\left(Z < \frac{52-50}{10}\right) = 0.37617$
- $P\left(X < \frac{t-50}{10}\right) - P(Z < 0.2) = 0.37617$, let $\frac{t-50}{10} = z_1$
- $P(X < z_1) - 0.57926 = 0.37617$
- $P(X < z_1) = 0.95543$
- From standard normal table, $z_1 = 1.7$.
- Hence $t = 67$

- **Problem 2:** Given a random variable X that follows normal distribution with $\mu = 40$ and $\sigma = 6$. Find the value of x that has
 - (a) 45% of the area to the left and
 - (b) 14% of the area to the right
- Ans:
- Given $X \sim N(40, 6)$.
- Convert $Z \sim N(0, 1)$ by $Z = \frac{x - 40}{6}$.

- (a) ; find x_1 , such that $P(X < x_1) = 0.45$
- $P\left(Z < \frac{x_1 - 40}{6}\right) = 0.45$, let $\frac{x_1 - 40}{6} = z_1$
- From standard normal table, $z_1 = -0.125$.
- Hence $x_1 = 39.25$.

- (b) ; find x_2 , such that $P(X > x_2) = 0.14$
- $P\left(Z > \frac{x_2 - 40}{6}\right) = 0.14$, let $\frac{x_2 - 40}{6} = z_2$
- $1 - P(Z < z_2) = 0.14 \rightarrow P(Z < z_2) = 0.86$
- From standard normal table, $z_2 = 1.08$.
- Hence $x_2 = 46.48$.

- **Problem 3:** The heights of 1000 students are approximately normally distributed with a mean of 174.5 centimeters and a standard deviation of 6.9 centimeters. How many students would expect to have heights,
 - (a) less than 160 cm
 - (b) 171.5 cm to 182 cm
 - (c) equal to 175 cm
 - (d) greater than or equal to 188 cm.

- **Problem 4:** A lawyer commutes daily from his suburban home to his midtown office. The average time for a one-way trip is 24 minutes, with a standard deviation of 3.8 minutes. Assume the distribution of trip times to be normally distributed.
 - (a) What is the probability that a trip will take at least 1/2 hour?
 - (b) If the office opens at 9:00 A.M. and the lawyer leaves his house at 8:45 A.M. daily, what percentage of the time is he late for work?
- (c) If he leaves the house at 8:35 A.M. and coffee is served at the office from 8:50 A.M. until 9:00 A.M., what is the probability that he misses coffee?
- (d) Find the length of time above which we find the slowest 15% of the trips.
- (e) Find the probability that 2 of the next 3 trips will take at least 1/2 hour.

- Let X represents the trip times from lawyer's suburban home to his midtown office normally distributed with the mean 24 minutes and a standard deviation of 3.8 minutes.

a)

Lets find the probability that a trip will take at least 1/2 hour.

$$\begin{aligned}P(X \geq 30) &= 1 - P(X < 30) \\&= 1 - P\left(\frac{X - \mu}{\sigma} < \frac{30 - 24}{3.8}\right) \\&= 1 - P(Z < 1.58) \\&= 1 - 0.9429 \\&= 0.0571\end{aligned}$$

b)

Lets find what percentage of the time is lawyer late for work if the office opens at 9:00 A.M. and the he leaves his house at 8:45 A.M. daily.

$$\begin{aligned}P(X > 15) &= 1 - P(X \leq 15) \\&= 1 - P\left(\frac{X - \mu}{\sigma} \leq \frac{15 - 24}{3.8}\right) \\&= 1 - P(Z \leq -2.37) \\&= 1 - 0.0089 \\&= 0.9911 = 99.11\%\end{aligned}$$

c)

Now lets calculate the probability that he misses coffee, If he leaves the house at 8: 35 A.M. and coffee is served at the office from 8: 50 A.M. until 9: 00 AM.

$$\begin{aligned}P(X > 25) &= 1 - P(X \leq 25) \\&= 1 - P\left(\frac{X - \mu}{\sigma} \leq \frac{25 - 24}{3.8}\right) \\&= 1 - P(Z \leq 0.23) \\&= 1 - 0.6026 \\&= 0.3974\end{aligned}$$

d)

Lets find the length of time above which we find the slowest 15% of the trips.

$$P(X > x) = 0.15$$

$$1 - P(X \leq x) = 0.15$$

$$P(X \leq x) = 1 - 0.15$$

$$P(X \leq x) = 0.85$$

$$P\left(Z \leq \frac{x - \mu}{\sigma}\right) = 0.85$$

$$P(Z < 1.04) = 0.85$$

$$\frac{x - \mu}{\sigma} = 1.04$$

$$x = 1.04 \cdot \sigma + \mu$$

$$x = 1.04 \cdot 3.8 + 24$$

$$x = 3.952 + 24$$

$$x = 27.952$$

e)

Lets determine the probability that 2 of the next 3 trips will take at least 1/2 hour.

$$\begin{aligned}P(X > 30) &= 1 - P(X \leq 30) \\&= 1 - P\left(\frac{X - \mu}{\sigma} < \frac{30 - 24}{3.8}\right) \\&= 1 - P(Z < 1.58) \\&= 1 - 0.9429 \\&= 0.0571\end{aligned}$$

Therefore, X has binomial distribution with parameters $n = 3$ and $p = 0.0571$

The probability mass function of X is:

$$\begin{aligned}P(X = x) &= \binom{n}{x} p^x (1 - p)^{n-x} \\&= \binom{3}{x} (0.0571)^x (1 - 0.0571)^{3-x} \quad x = 0, 1, 2, 3\end{aligned}$$

Therefore the probability that 2 of the next 3 trips will take at least 1/2 hour is:

$$\begin{aligned}P(X = 2) &= \binom{3}{2} (0.0571)^2 (1 - 0.0571)^{3-2} \\&= 3 \cdot 0.00326 \cdot 0.9429 \\&= \mathbf{0.0092}\end{aligned}$$

- **Problem 5:** If a set of observations is normally distributed. What percentage of these differ from mean by
 - (a) more than 1.3σ
 - (b) less than 0.52σ
 - (c) between 1.52σ and 2.3σ

- **Problem 6:** The average life of a certain type of small motor is 10 years with a standard deviation of 2 years. The manufacturer replaces free all motors that fail while under guarantee. If she is willing to replace only 3% of the motors that fail, how long a guarantee should be offered? Assume that the lifetime of a motor follows a normal distribution

- Let random variable X represent the lifetime of a motor that follows a normal distribution with mean 10 years and the standard deviation of 2 years.

Lets calculate how long a guarantee should be offered if she is willing to replace only 3% of the motors that fail.

$$P(X \leq x) = P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = P\left(Z \leq \frac{x - \mu}{\sigma}\right) = 0.03$$

$$_2 \quad P(Z \leq -1.88) = 0.03$$

Therefore,

$$\frac{x - \mu}{\sigma} = -1.88$$

$$x = -1.88 \cdot \sigma + \mu$$

$$x = -1.88 \cdot 2 + 10$$

$$x = 6.24$$

- **Problem 7:** A soft-drink machine is regulated so that it discharges an average of 200 milliliters per cup. If the amount of drink is normally distributed with a standard deviation equal to 15 milliliters,
 - (a) what fraction of the cups will contain more than 224 milliliters?
 - (b) what is the probability that a cup contains between 191 and 209 milliliters?
 - (c) how many cups will probably overflow if 230-milliliter cups are used for the next 1000 drinks?
 - (d) below what value do we get the smallest 25% of the drinks?

- Let X represent the amount of drink distributed.
- The mean is $\mu = 200$ milliliters per cup.
- The standard deviation $\sigma = 15$ milliliters per cup.

a)

Lets calculate what fraction of the cups will contain more than 224 milliliters.

$$z = \frac{x - \mu}{\sigma} = \frac{224 - 200}{15} = 1.6$$

Therefore,

$$\begin{aligned} P(X > 224) &= P(Z > 1.6) \\ &= 1 - P(Z < 1.6) \\ &= 1 - 0.9452 \\ &= 0.0548 \end{aligned}$$

b)

Now, let's find the probability that a cup contains between 191 and 209 milliliters.

$$z = \frac{x_1 - \mu}{\sigma} = \frac{191 - 200}{15} = -0.6$$

$$z = \frac{x_2 - \mu}{\sigma} = \frac{209 - 200}{15} = 0.6$$

Therefore,

$$\begin{aligned} P(-0.6 < X < 0.6) &= P(-0.6 < Z < 0.6) \\ &= P(Z < 0.6) - P(Z < -0.6) \\ &= 0.7257 - 0.2743 \\ &= \mathbf{0.4514} \end{aligned}$$

d)

Now, lets find below what value do we get the smallest 25% of the drinks.

$$P(X < x) = 0.25$$

$$P\left(Z < \frac{x - \mu}{\sigma}\right) = 0.25$$

$$P(Z < -0.68) = 0.25$$

$$\frac{x - \mu}{\sigma} = -0.68$$

$$x = -0.68 \cdot \sigma + \mu$$

$$x = -0.67 \cdot 15 + 200$$

$$x = 189.95$$

- **Problem 9:** The time taken for Alex Hatz to come to school is normally distributed with mean 30 min and standard deviation 7 min.
- (a) Given that Alex leaves his home at 8:05, find the probability that he will be late at school (i.e. he arrives after 8:30, when lessons start!)
- (b) If Alex wishes to be on time (i.e. before 8:30) at least 90% of the days. Find the time he should leave home every day.
- The time taken for another student, called Ioannis, to come to school is also normally distributed with mean 30 min and standard deviation σ .
- (c) Given that Ioannis leaves his home at 8:02, and is late 60% of the morning classes, find the standard deviation σ .
- (d) For a particular 5-days week, Ioannis leaves his home at 7:59. Find the probability that he is on time at least twice.

- **Problem 10:** If X and Y are two independent normal random variable such that $X \sim N(100, 11)$ and $Y \sim N(150, 7)$.
- (a) Find the mean and standard deviation of the random variable $U = 2X - 3Y$.
- (b) Find $P(U < -234)$

- **Problem 11:** On average, cows produce 49 units of milk per day with standard deviation 5.87 units, whereas Jersey cows produce 44.8 units of milk per day with standard deviation 5.12 units. For each breed, the production of milk can be modelled by a normal distribution.
- A) Find the probability that a randomly selected Ayrshire cow will produce more than 50 units of milk daily.
- B) Find the probability that a randomly selected Jersey cow will produce more milk than a randomly selected Ayrshire cow.
- C) A dairy farmer has 25 Jersey cows. Find the probability that the daily production for this small herd exceeds 46 units per cow per day.
- D) A neighboring farmer has 15 Ayrshire cows. Find the probability that this herd produces at least 4 units per cow per day more than the Jersey herd.

